

Positivity of the Hartree term of EDUS

why ρ_0 can be unstable with the mean field, and the options `hartree_centers` and `hartree_cutoff_factor`

EDUS development notes

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Abstract

In EDUS the Hartree term is the interaction between point charges sitting at the Wannier centers, saturated below a cutoff distance. With several Wannier functions per atom (monolayer MoS₂: five Mo d and six S p) the resulting Hartree matrix has negative eigenvalues: some charge-neutral redistributions of the occupations *lower* the Hartree energy, which is impossible for a physical Coulomb repulsion. Then ρ_0 is not a minimum of the mean-field energy, and the propagation without any perturbation runs away in a few femtoseconds (it is not a time-step problem). These notes explain the mechanism and the two input options that fix it, and the check printed in the recap.

1 The Hartree term

`MeanField::initialize` builds, from the bare interaction of `ModelCoulomb` (`vcoul3d`, $\varepsilon = 1$),

$$H_{ij} = \sum_{\mathbf{R}} V(|\boldsymbol{\tau}_i - \mathbf{R} - \boldsymbol{\tau}_j|), \quad V(r) = \frac{s}{\max(r, d_c)}, \quad (1)$$

with $\boldsymbol{\tau}_i$ the center of the Wannier function i , the sum over the $\mathbf{R}$ grid of the simulation, $s = 2$ and d_c the saturation (cutoff) distance. The self-energy is diagonal and local, $\Sigma_{ii}^H = \sum_j H_{ij} \Delta n_j$ with $\Delta n_j = \rho_{jj}(\mathbf{R} = 0) - \rho_{0,jj}(\mathbf{R} = 0)$, and its energy

$$E_H = \frac{1}{2} \Delta n^T H \Delta n. \quad (2)$$

Until now the cutoff was the smallest distance between two centers on different sites, $d_c = d_{\min} = \min_{|\boldsymbol{\tau}_i - \boldsymbol{\tau}_j| > 0.1} |\boldsymbol{\tau}_i - \boldsymbol{\tau}_j|$ (centers closer than 0.1 bohr are considered the same site). So the on-site interaction is $V(d_{\min})$: the interaction between the two closest centers.

2 Why negative eigenvalues make ρ_0 unstable

The number of electrons is conserved, $\sum_j \Delta n_j = 0$: only the eigenvalues of H on this charge-neutral subspace matter (the projection $P = 1 - \mathbf{1}\mathbf{1}^T/n$ removes the large constant coming from the long-range sum). For a physical charge density the Hartree energy is positive,

$$E_H = \frac{1}{2} \iint \frac{\delta n(\mathbf{r}) \delta n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{2} \int \frac{d^3 q}{(2\pi)^3} \frac{4\pi}{q^2} |\delta n(\mathbf{q})|^2 \geq 0,$$

because the Fourier transform of $1/r$ is positive. A matrix like (1) is a valid Coulomb interaction only if PHP is positive semidefinite. If it has a negative eigenvalue, moving charge along that eigenvector lowers E_H . Whether ρ_0 becomes unstable depends on the competition with the cost in band energy (roughly the gap, through χ_0^{-1} in the static response): when the negative Hartree energy wins, ρ_0 is a saddle point of the mean-field energy and the linearized dynamics has growing modes. Such an instability does not depend on ε or on the screened exchange (the bare interaction is not screened in EDUS), it is there in RPA, and it is physical for the model: the total energy is conserved while the band energy grows. A negative eigenvalue is a defect of the model even when the gap hides it: the static screening of the model is then wrong.

Observed on MoS₂ (PBE, 11 Wannier functions, 6×6, HSEX, no laser, no phonons): $\text{min eig}(PHP) = -2.22$ Ha. The band energy goes from 10^{-17} to 0.3 Ha in 1–2 fs while the mean-field energy goes to -0.3 Ha, with the total energy conserved to 10^{-8} Ha, then NaN. Same with $\varepsilon = 5, 10$, with RPA, on 12×12, and at $\Delta t = 0.001$ fs. hBN (one orbital per atom) is not affected.

3 Why the cutoff at $d_{\min}$ is not physical

A point charge with a cutoff stands for a charge spread over a Wannier function. Its self-interaction must be clearly larger than the interaction with the neighbours; with $d_c = d_{\min}$ it is not.

Two sites per cell (hBN). For $\Delta n = (1, -1)$: $2E_H = H_{BB} + H_{NN} - 2H_{BN}$. Both H_{BB} and H_{BN} are lattice sums over the whole crystal, and the difference comes from the closest terms: in H_{BB} the $R = 0$ term is the on-site $V(d_c)$, in H_{BN} there are the three nearest neighbours at d_{BN} . With $d_c = d_{\min} = d_{BN}$ the on-site term equals one neighbour, while B has three: $H_{BB} = 17.86 < H_{BN} = 18.23$ Ha (12×12, hBN model of the tests) and $\text{min eig}(PHP) = -0.37$ Ha. hBN is nevertheless stable in the tests because its gap (7.25 eV) is large enough; with the DFT gap (~ 4.7 eV) the mean-field propagation of hBN was found unstable, which may be the same mechanism. With $d_c = d_{BN}/2$ the eigenvalue is positive.

Three sites (MoS₂ with the orbitals of an atom at the same position, see below). The grouped matrix has on site $U = 6.45$ Ha, Mo–S $V_{\text{MoS}} = 6.25$ Ha, S–S $V_{\text{SS}} = 5.42$ Ha (lattice sums included; the S are farther from each other, 5.9 bohr, than from Mo, 4.6 bohr). Moving two electrons from Mo, one to each S, $\Delta n = (2, -1, -1)$:

$$2E_H = U(4+1+1)+2[(2)(-1)V_{\text{MoS}}+(-1)(-1)V_{\text{SS}}] = 6U-8V_{\text{MoS}}+2V_{\text{SS}} = 38.7-50.0+10.8 = -0.46 \text{ Ha.}$$

The self-repulsion $6U$ is smaller than the Mo–S attraction gained, $8V_{\text{MoS}}$, because the $R = 0$ term of U is the nearest-neighbour value $V(d_{\text{MoS}})$ while each atom has several neighbours in the lattice sums; and the two negative charges repel each other only through the small V_{SS} .

Several centers on one atom (the actual Wannier centers of MoS₂). The five Mo d centers differ by up to 0.43 bohr, the closest two by $d_{\min} = 0.13$ bohr. Some on-site pairs get the saturated value $s/0.13$, others s/r with r up to 0.43 bohr: entries from 10.6 to 21.7 Ha within the same atom. Moving charge between the d orbitals of Mo then changes E_H by large amounts of both signs: $\text{min eig}(PHP) = -2.22$ Ha. The differences between the centers are a property of the Wannier gauge, not physics: the five d functions are centered on the same atom.

4 The fix

Two options of the input (top level, with `coulomb`), used only for the Hartree term; the screened exchange keeps the actual Wannier centers.

- `hartree_centers = "atoms"` (default `"wannier"`): centers closer than `hartree_center_tolerance` (default 0.6 Å, directly or through other centers) are grouped, and each one is moved to the mean of its group. All the orbitals of an atom then have the same on-site interaction.
- `hartree_cutoff_factor` (default 1): $d_c = f d_{\min}$. With $f = 1/2$ and grouped centers the on-site interaction is $V(d_{\text{NN}}/2) = 2V(d_{\text{NN}})$, twice the nearest-neighbour one; for MoS₂ $U = s/2.28$ bohr, i.e. 0.44 Ha = 12 eV per spin, a reasonable bare U for d orbitals.

centers	cutoff	min eig(PHP) (Ha), 6×6	12×12
Wannier	$d_{\min}$ (0.13 bohr)	-2.22	-2.22
Wannier	$d_{\min}/2$	$-4 \cdot 10^{-4}$	$-2 \cdot 10^{-4}$
Wannier	$d_{\min}/4$	$+4 \cdot 10^{-5}$	
atoms	$d_{\min}$ (4.57 bohr)	-0.30	-0.42
atoms	$d_{\min}/2$	0 (10^{-15})	0

For hBN (one center per atom) $f = 1/2$ alone gives a positive eigenvalue. With atoms and $f = 1/2$ the nonzero eigenvalues are 1.49 and 4.39 Ha; the zero ones are the redistributions of charge among the orbitals of the same atom, which a monopole Hartree term does not see. Without grouping a smaller cutoff also removes the negative eigenvalues, but at the price of an unphysical on-site interaction ($s/0.03$ bohr, hundreds of eV). MoS₂ with `hartree_centers = "atoms"` and `hartree_cutoff_factor = 0.5` stays at $E_{\text{band}} \sim 10^{-16}$ Ha for 40 fs without perturbation, also with $\Delta t = 0.04$ fs.

Limits. A cutoff on point charges does not guarantee positivity for every geometry. A kernel with a positive Fourier transform would, e.g. the Ohno form $V(r) = 1/\sqrt{r^2 + a^2}$ (Fourier transform $\propto aK_1(qa)/q > 0$), with $V(0) = 1/a$ the bare on-site U (not implemented; for MoS₂ it gives $\text{min eig} \sim -10^{-5}$ Ha with $a = 1\text{--}3$ bohr, the residual coming from the finite sum over $\mathbf{R}$).

5 Check in the recap

The `MEAN FIELD` section of the output prints the Hartree centers, the cutoff distance and $\text{min eig}(PHP)$ (“Hartree: min eigenvalue”), with a warning if it is below -10^{-8} Ha. A negative value means that the Hartree term is not a physical repulsion; ρ_0 is unstable, whatever the time step, when it exceeds the band cost. All the HSEX regression tests on hBN have $-0.37/-0.38$ Ha with the defaults (stable thanks to the gap); Si, GaAs and LiF have 0. The defaults are kept (`"wannier", 1`) so that the existing results do not change.